NSBM Green University

SE102.3 Web Based Application Development

Web Design - Repeat Coursework

PROJECT REPORT

Daily Pulse News

Online News Website with Admin Features

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1. Introduction

Daily Pulse News is a dynamic online news portal. Readers can browse the latest stories, filter them by category, search for topics and leave comments. Editors and administrators use a protected admin panel to publish and manage the content. The website is backed by a MySQL database and supports all DML operations (INSERT, SELECT, UPDATE and DELETE).

The project satisfies the coursework requirements: a planned UI, a website with more than four main pages, a database with all DML operations, an admin section, and GitHub for version control and contribution tracking.

2. Proposal

2.1 Problem statement

Small news organisations need a simple and affordable way to publish stories online and manage them without editing code.

2.2 Objectives

1. Provide a clean, responsive public news website.
2. Provide an admin panel for managing articles, categories, comments, messages and users.
3. Store all data in a relational database.
4. Use GitHub for version control and to show contributions.

2.3 Scope

Public pages: Home, Category, Article, Search, About Us and Contact Us.

Admin pages: Login, Dashboard, Articles, Categories, Comments, Messages and Users.

2.4 Technologies

Area	Technology
Back end	PHP 8 with PDO
Database	MySQL / MariaDB
Front end	HTML5, CSS3
Server	XAMPP (Apache)
Version control	Git and GitHub
Editor	Visual Studio Code

2.5 User roles

- **Visitor -** reads, searches, comments and sends messages.
- **Editor -** logs in and manages articles, categories, comments and messages.
- **Admin -** everything an editor can do, plus user management.

3. UI Plan (Web Design)

3.1 Layout and style

Every page shares a consistent header (logo and search box), a dark navigation bar with category links, a main content area and a footer.

Element	Choice
Accent colour	Red #C8102E
Navigation / footer	Charcoal #1B1F27
Panels	Light grey #F3F4F6
Typography	Segoe UI / Arial; section titles have a red left border
Responsive	Three-column grids collapse to one column below 800px

3.2 Wireframe - Home page

```
+-------------------------------------------------+
| Logo                        [ Search ........ ][Go]|
| Home World Sports Tech Business Ent. About Contact  |
+-------------------------------------------------+
| TOP STORIES                                       |
| [ card ]        [ card ]        [ card ]          |
+-----------------------------------+---------------+
| LATEST NEWS                       | MOST READ     |
| [ card ]  [ card ]                | 1. ...        |
| [ card ]  [ card ]                | 2. ...        |
+-----------------------------------+---------------+
| Footer                                            |
+-------------------------------------------------+
```

3.3 Site map

```
Home
 |- Category (World, Sports, Technology, Business, Entertainment)
 |    `- Article (+ comments)
 |- Search results
 |- About Us
 |- Contact Us
 `- Admin Login
      `- Dashboard
           |- Articles   (list, add, edit, delete)
           |- Categories (add, edit, delete)
           |- Comments   (delete)
           |- Messages   (delete)
           `- Users      (add, change role, delete)
```

4. Website GUI

This section shows the implemented pages. The public site has six pages and the admin panel has seven.

4.1 Public website

Home page

Shows Top Stories (featured articles), the Latest News grid and a Most Read sidebar ordered by view count.

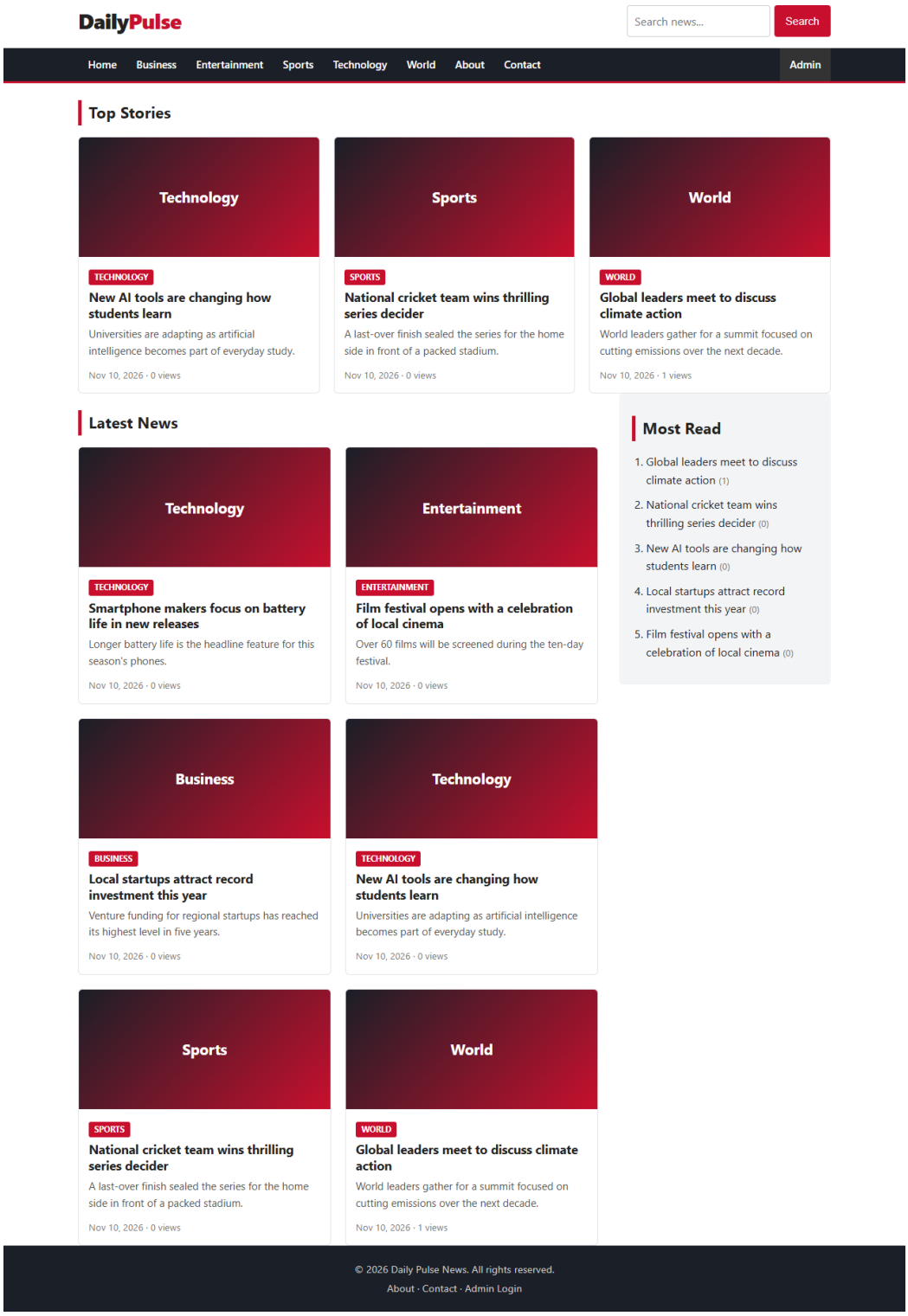


Figure 1: Home page

Category page

Lists every article that belongs to the selected category.

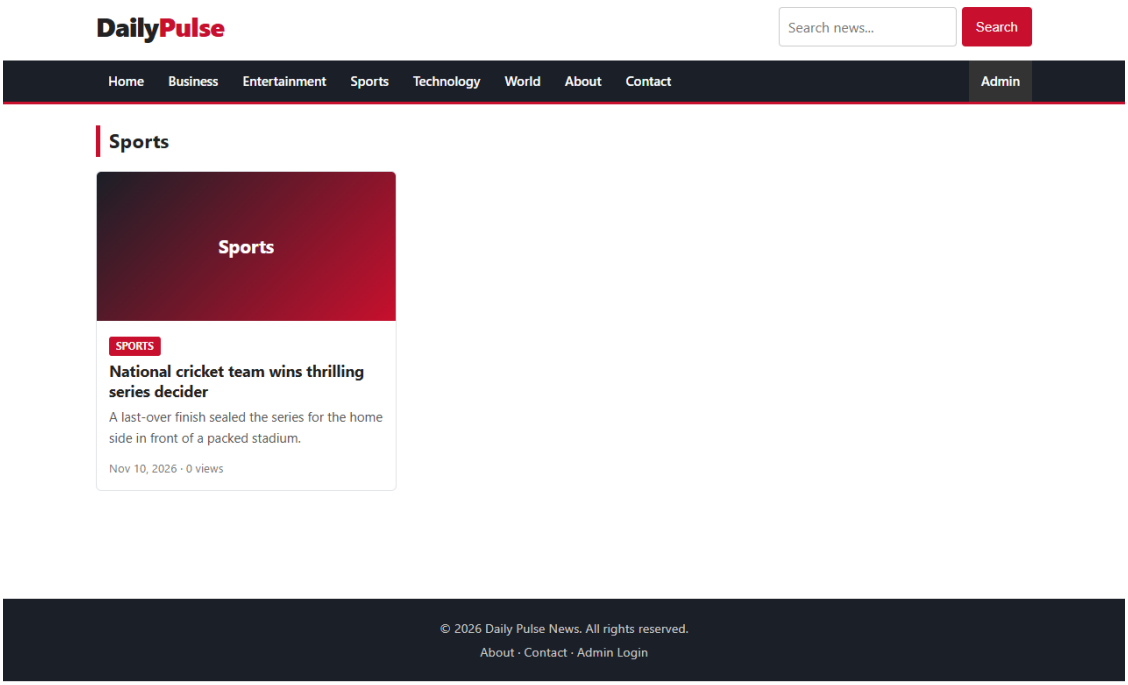


Figure 2: Category page (Sports)

Article page

Shows the full story, author, date and view count. Visitors can post comments.

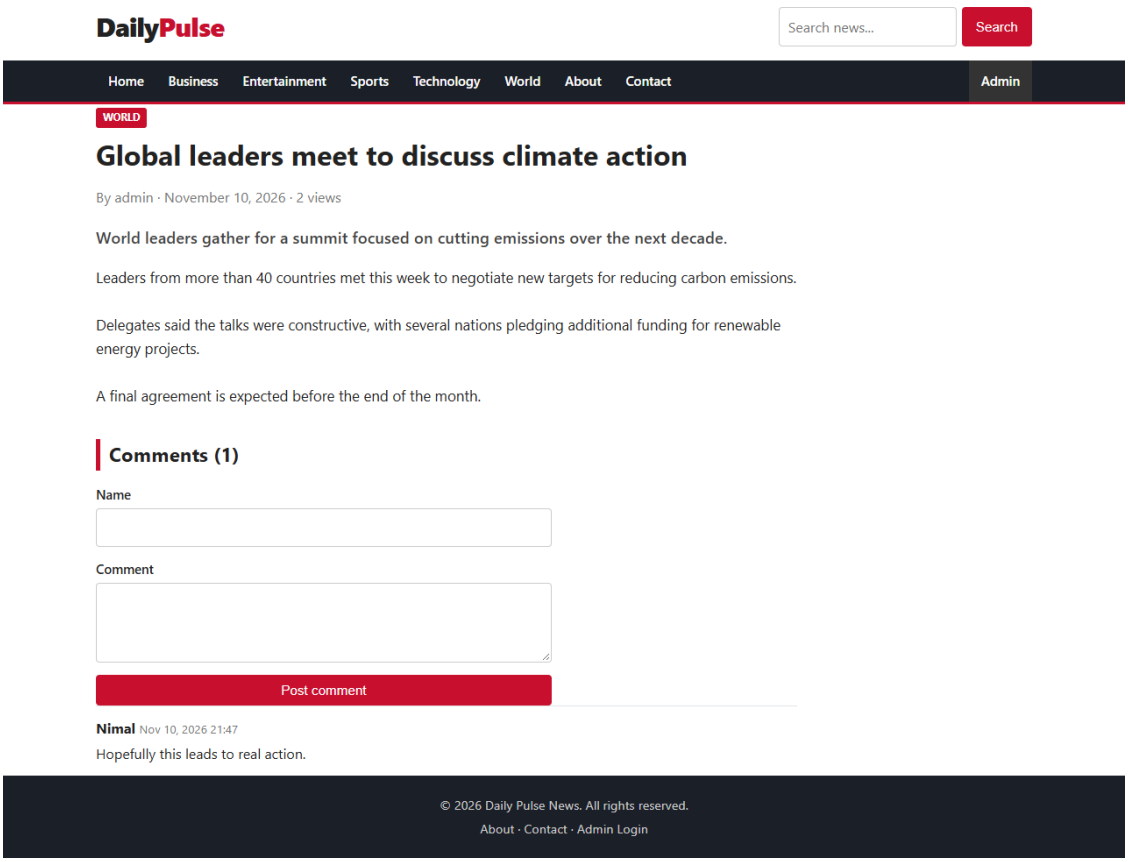


Figure 3: Article page with comments

Search page

Searches the title, summary and content of all articles.

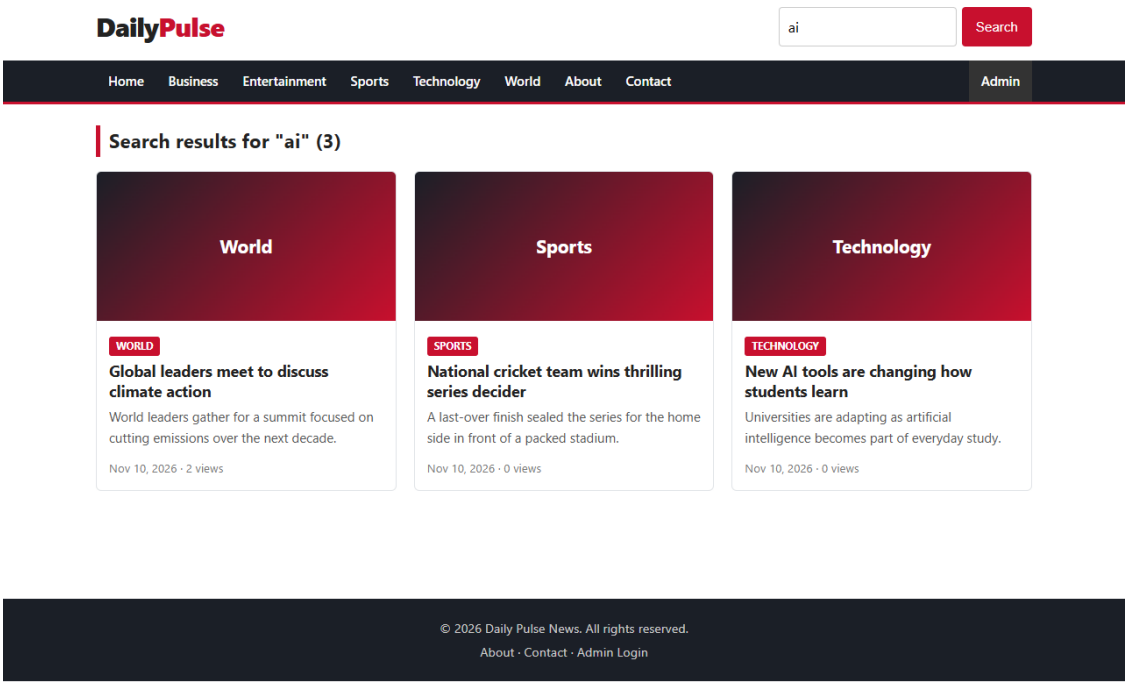


Figure 4: Search results

About Us page

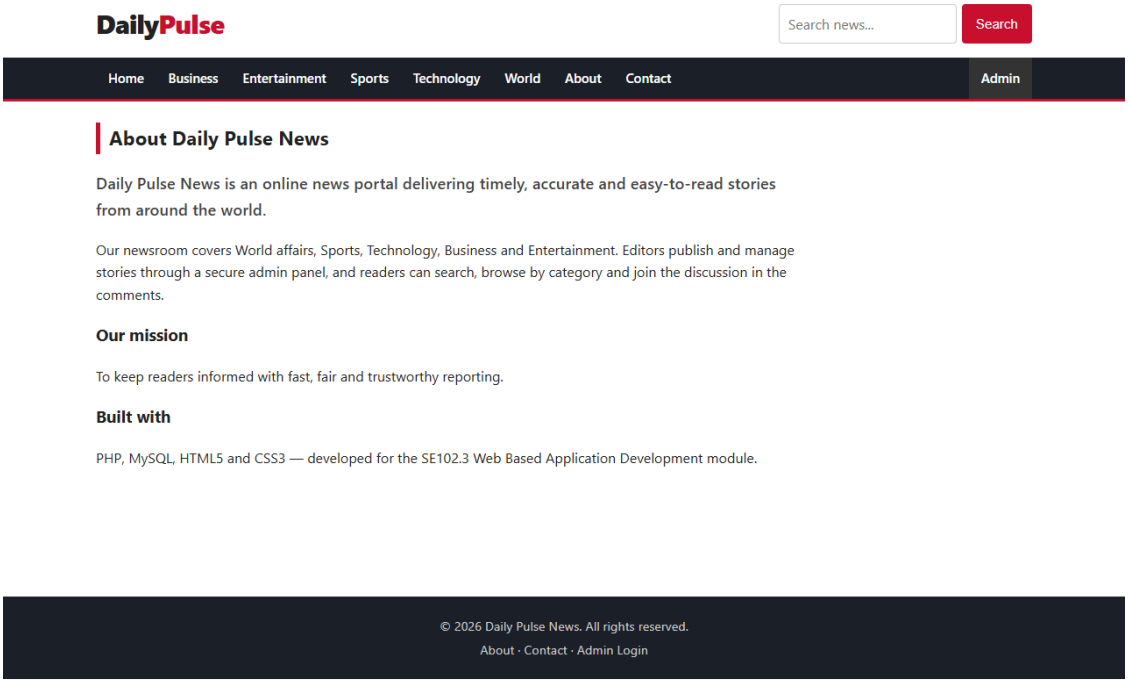


Figure 5: About Us page

Contact Us page

Validated form. Submitted messages are stored in the messages table.

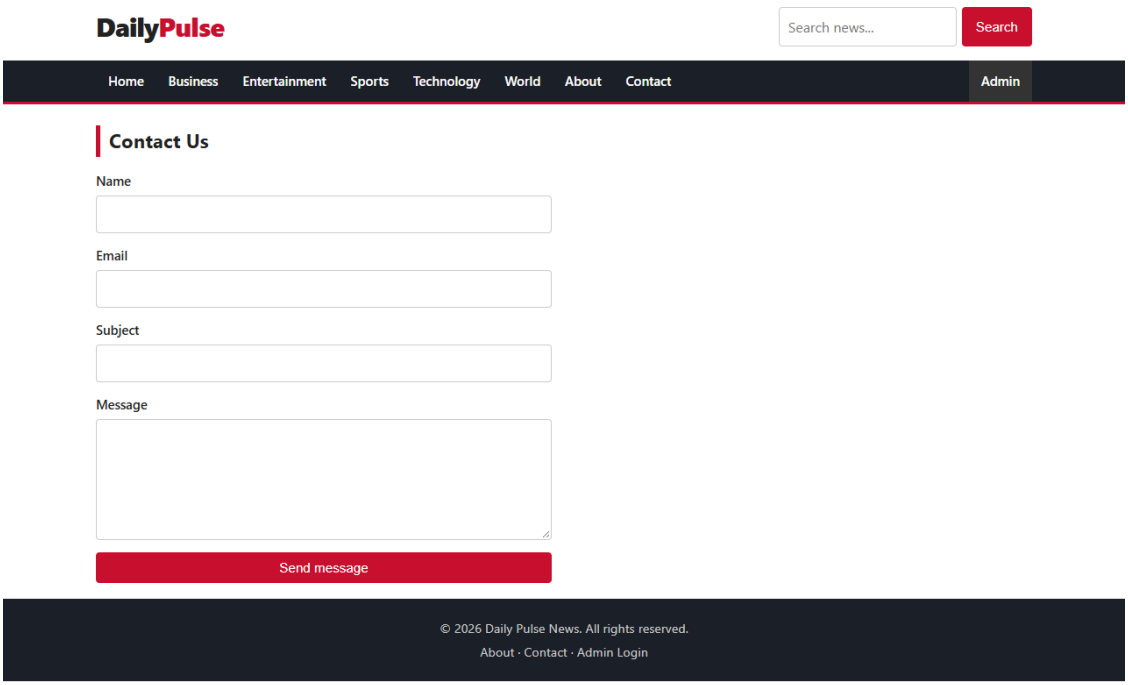


Figure 6: Contact Us page

4.2 Admin panel

Admin login

Staff sign in with a username and password. Passwords are stored as bcrypt hashes.

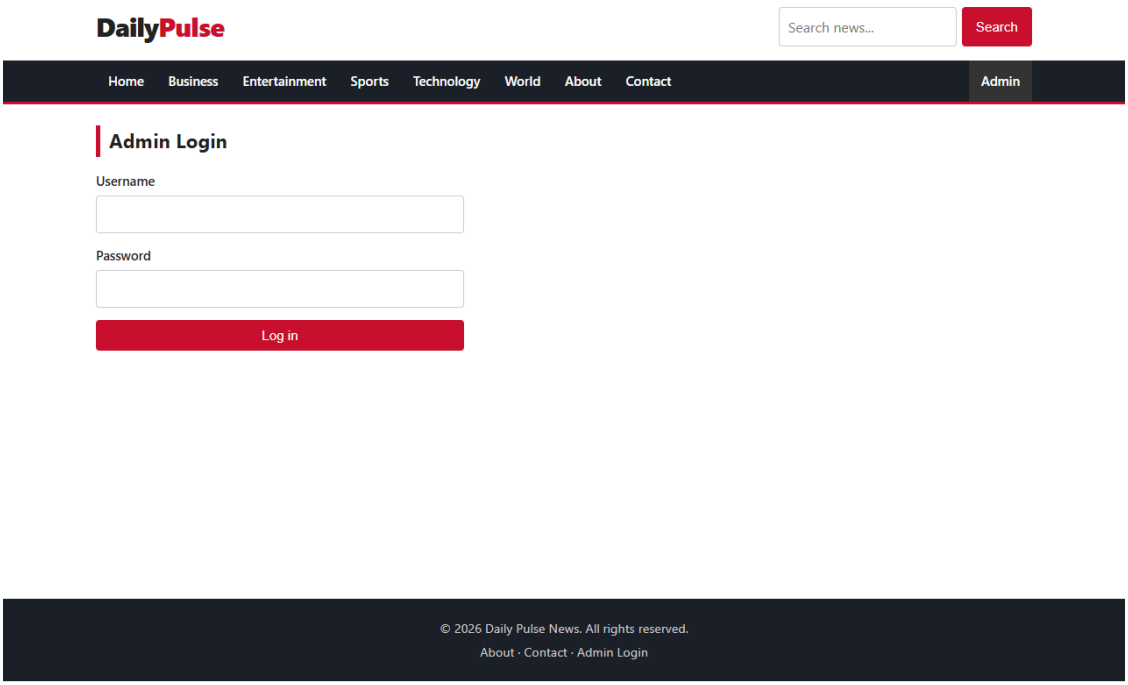


Figure 7: Admin login

Dashboard

Displays totals for articles, categories, comments, messages and users, and the five most recent articles.

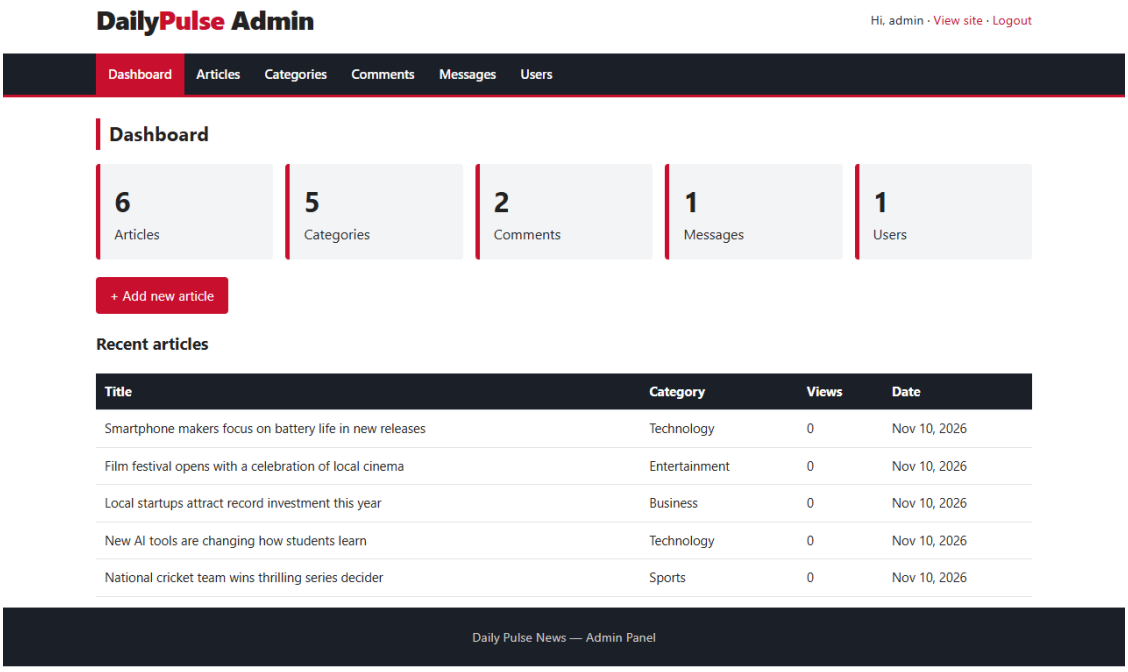


Figure 8: Admin dashboard

Manage articles

Table of all articles with View, Edit and Delete actions.

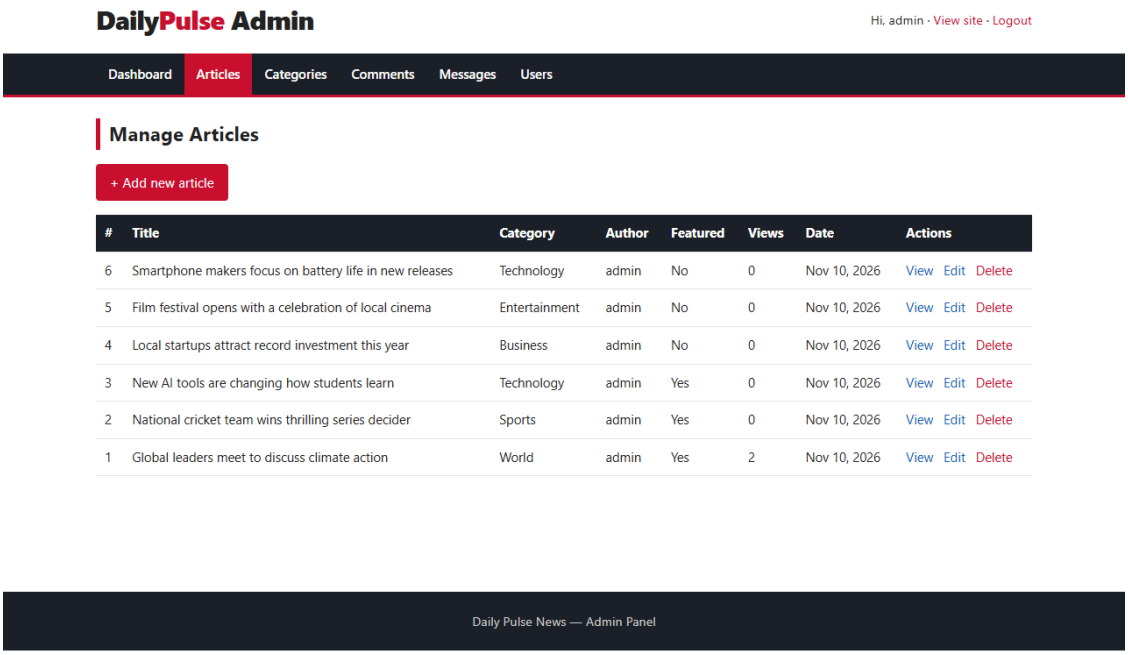


Figure 9: Manage articles

Add / edit article

Form with title, category, summary, content, optional image upload and a Top Stories option.

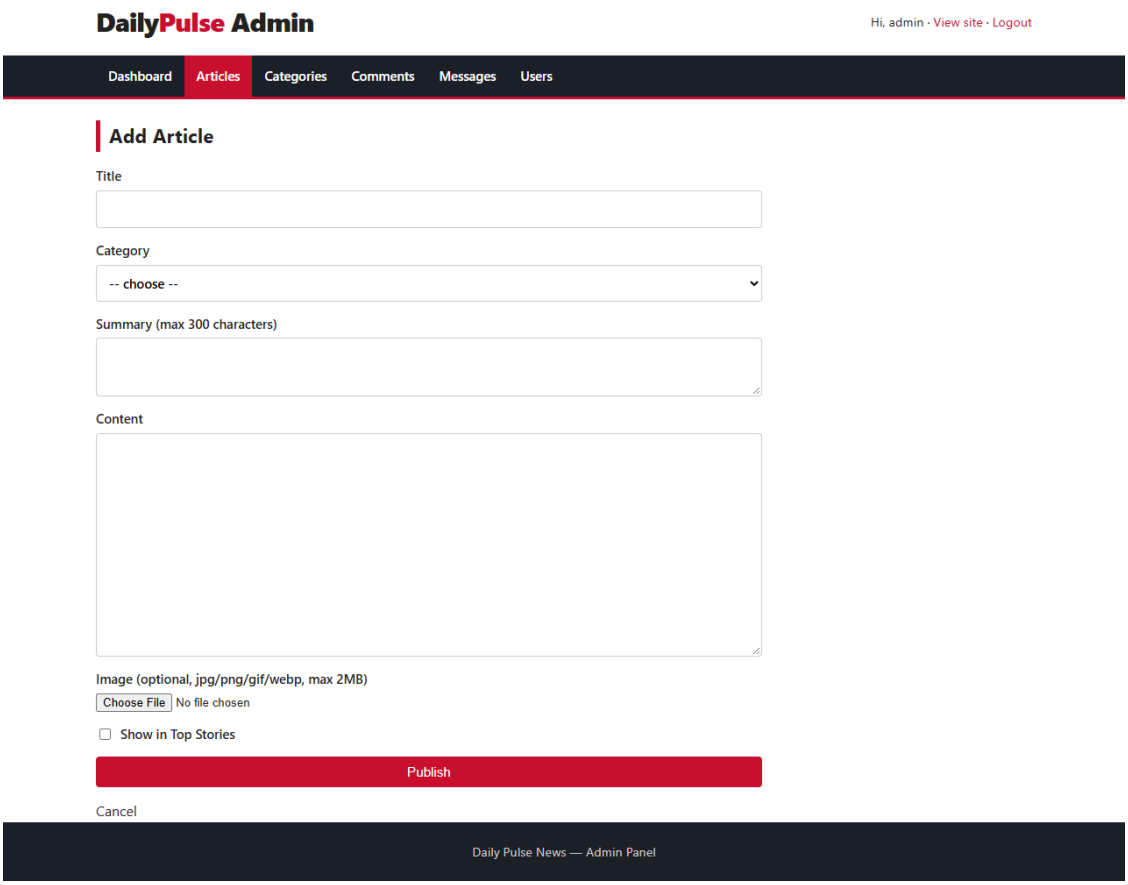


Figure 10: Add article form

Manage categories

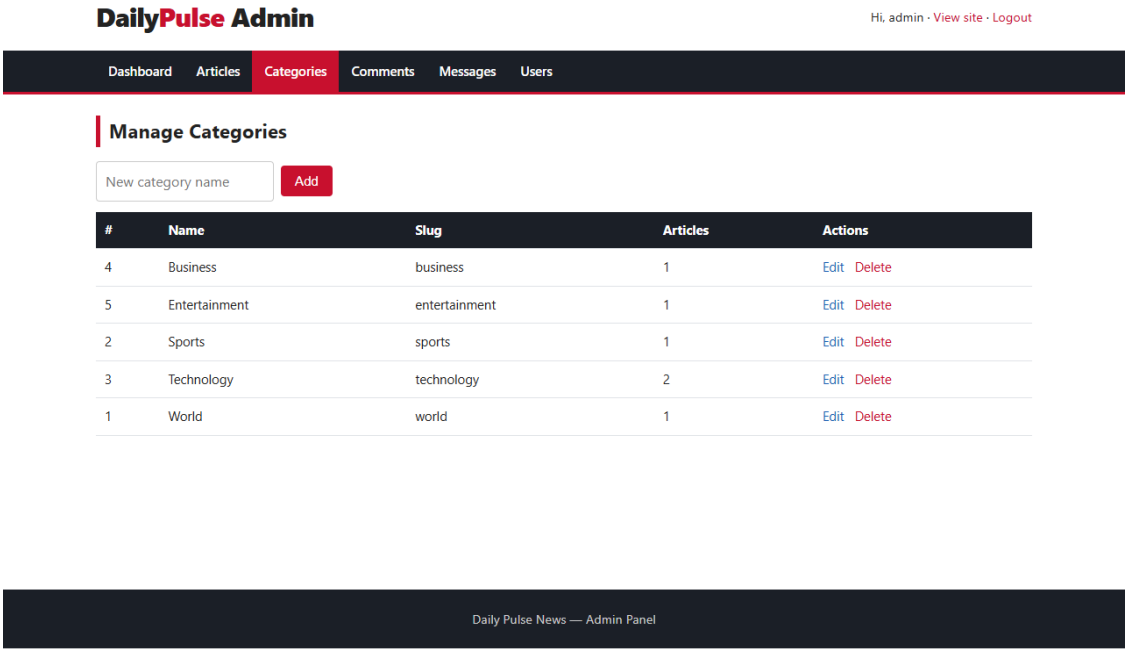


Figure 11: Manage categories

Moderate comments

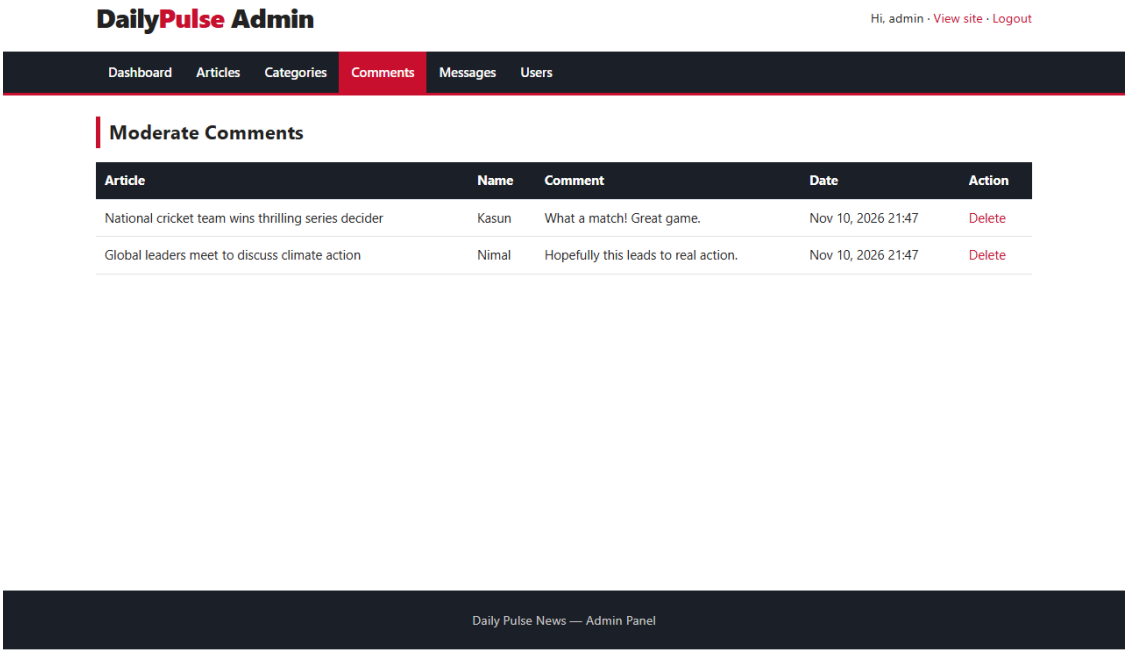


Figure 12: Moderate comments

Contact messages

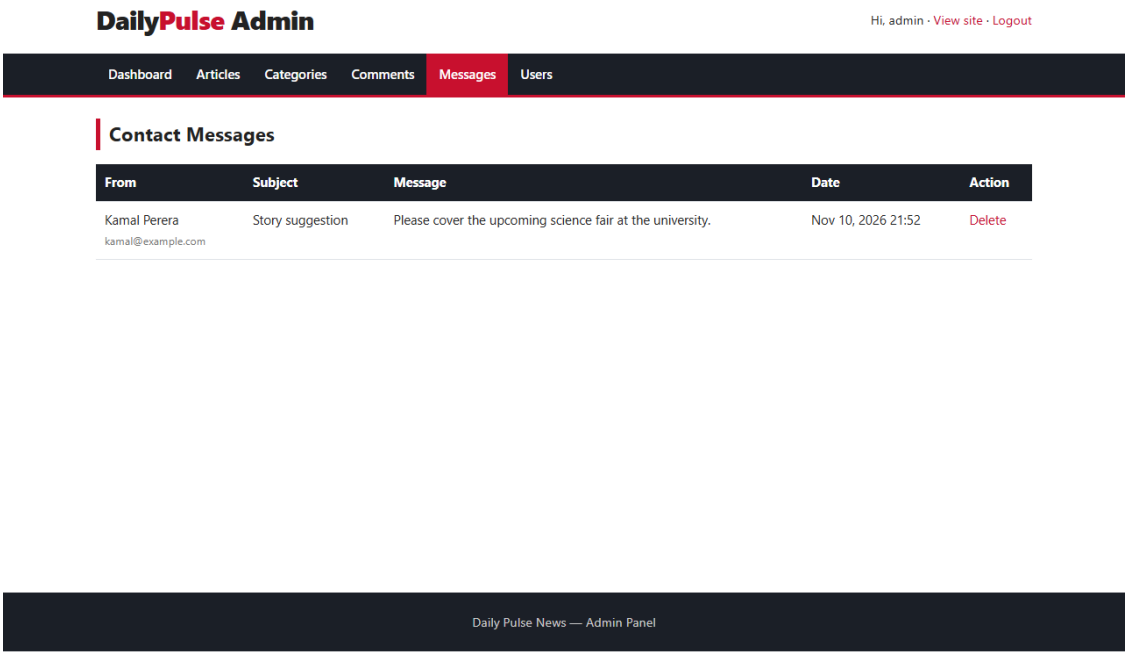


Figure 13: Contact messages

Manage users

Available to administrators only. Roles can be changed and users removed (the signed-in user cannot delete or demote themselves).

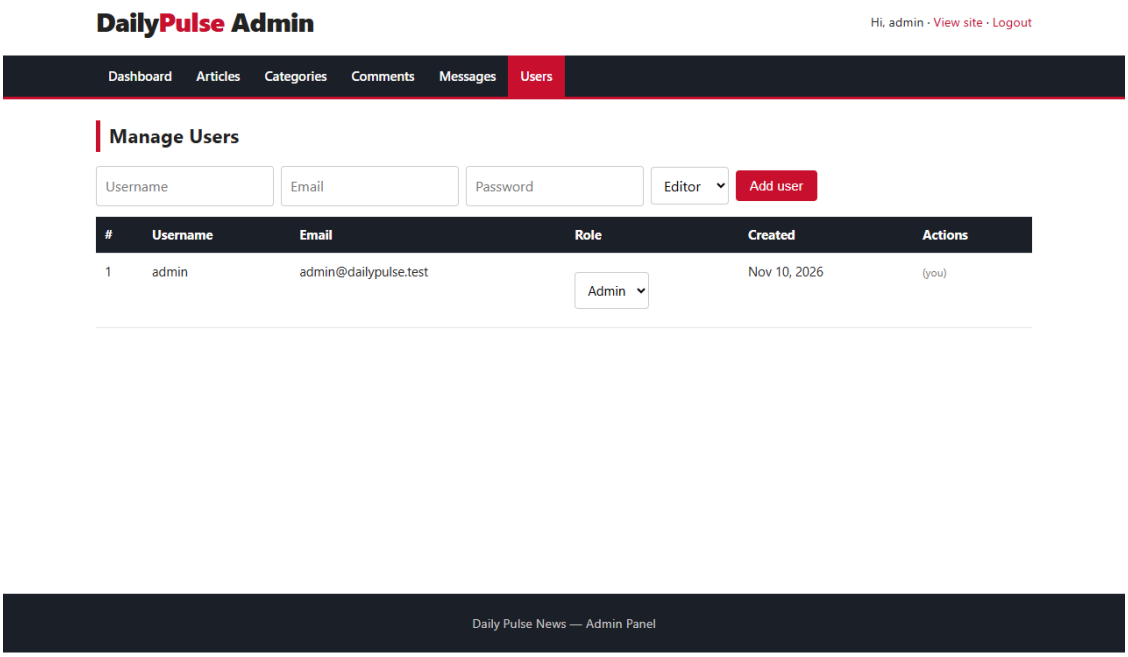


Figure 14: Manage users

5. Database Design

Database name: news_portal (engine InnoDB, character set utf8mb4). The full script is in database/schema.sql.

5.1 Tables

users

Column	Type	Notes
id	INT, PK, AUTO_INCREMENT	
username	VARCHAR(50)	UNIQUE
email	VARCHAR(100)	UNIQUE
password	VARCHAR(255)	bcrypt hash
role	ENUM('admin','editor')	default editor
created_at	TIMESTAMP	

categories

Column	Type	Notes
id	INT, PK, AUTO_INCREMENT	
name	VARCHAR(60)	UNIQUE
slug	VARCHAR(80)	UNIQUE, used in URLs

articles

Column	Type	Notes
id	INT, PK, AUTO_INCREMENT	
title	VARCHAR(200)	
summary	VARCHAR(300)	
content	TEXT	
image	VARCHAR(255), NULL	file name in /uploads
category_id	INT, FK	categories(id), ON DELETE CASCADE
author_id	INT, FK	users(id), ON DELETE CASCADE
is_featured	TINYINT(1)	shown in Top Stories
views	INT	incremented on each view
created_at	TIMESTAMP	

comments

Column	Type	Notes
id	INT, PK, AUTO_INCREMENT	

Column	Type	Notes
article_id	INT, FK	articles(id), ON DELETE CASCADE
name	VARCHAR(80)	
body	TEXT	
created_at	TIMESTAMP	

messages

Column	Type	Notes
id	INT, PK, AUTO_INCREMENT	
name	VARCHAR(80)	
email	VARCHAR(100)	
subject	VARCHAR(150)	
message	TEXT	
created_at	TIMESTAMP	

5.2 Relationships

```
users 1 ----- * articles * ----- 1 categories
                    1
                    |
                    *
                comments

messages (stand-alone)
```

- One user writes many articles.
- One category contains many articles.
- One article has many comments.

5.3 DML operations

Operation	Example	File
INSERT	INSERT INTO articles (...) VALUES (?,...)	admin/article_form.php
INSERT	INSERT INTO comments ...	article.php
INSERT	INSERT INTO messages ...	contact.php
SELECT	SELECT a.*, c.name FROM articles a JOIN categories c ...	index.php, category.php
SELECT	... WHERE title LIKE ? OR summary LIKE ? OR content LIKE ?	search.php
UPDATE	UPDATE articles SET title=?, ... WHERE id=?	admin/article_form.php
UPDATE	UPDATE articles SET views = views + 1	article.php
DELETE	DELETE FROM articles WHERE id = ?	admin/articles.php
DELETE	categories, users, comments, messages	admin/*.php

6. Security

- Passwords are stored with password_hash() and verified with password_verify().
- All SQL uses PDO prepared statements, which prevents SQL injection.
- All output is escaped with htmlspecialchars(), which prevents XSS.
- A CSRF token protects every form that changes data.
- Admin pages require a login session; user management is limited to the admin role.
- Image uploads are restricted by file type and size and saved with random file names.

7. GitHub and Contributions

Repository: https://github.com/kariyawasamp679-create/news-portal-web-app

The work was committed in stages so that the history shows how the project was developed:

No.	Commit message
1	Add database schema, sample data and configuration
2	Add public pages: home, category, article, search, about, contact and login
3	Add admin panel with article, category, comment, message and user management
4	Add responsive stylesheet
5	Add README and project report

The Insights > Contributors page and the commit history of the repository show the contribution timeline.

8. Testing

Test	Result
All public pages load without errors	Pass
Admin login with correct password	Pass
Admin login with wrong password is rejected	Pass
Admin pages redirect to login when not signed in	Pass
Create, edit and delete an article	Pass
Add a category	Pass
Post a comment	Pass
Send a contact message	Pass
Search returns matching articles	Pass

9. How to Run the Project

5. Copy the project folder to C:\xampp\htdocs\news-portal.
6. Start Apache and MySQL in the XAMPP Control Panel.
7. Open phpMyAdmin, choose Import, select database/schema.sql and click Go.

8. Open http://localhost/news-portal/ in a browser.
9. Admin login: http://localhost/news-portal/login.php (username: admin, password: Admin@123).

10. Conclusion

The project delivers a complete, database-driven news website with a secure admin panel and full DML support, built and tracked on GitHub. Possible future improvements include a rich-text editor, article scheduling, pagination, and reader registration.